

The Yield Curve as an Economic Signal:

Predicting Recessions and Stock Market Performance

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Introduction

The world economy experiences cycles of expansion and contraction, and forecasting such cycles is crucial for financial analysts, investors, and policymakers. In the past few decades, the U.S. Treasury yield curve, and indeed the long-term (10-year) minus short-term (2-year) spread, has been a very reliable leading indicator. An inversion of the yield curve, i.e., when short-term yields are more significant than long-term yields, has historically resulted in economic recessions. This research examines if the yield spread has the potential to be a good leading indicator of recession events and influence stock market behaviour, as measured by S&P 500 returns. With the assistance of different sources of data from Federal Reserve Economic Data (FRED) to Yahoo Finance stock data and sound statistical methods, this research tries to locate the dynamics of said relationships. The driving impetus for this research is refining our knowledge of economic indicators to inform risk management and policy design decisions better. By extension, the study feeds into the wider body of research about economic prediction and the efficiency of financial markets.

Methods

In investigating the forecasting power of the U.S. Treasury yield curve to the letter, a multi-stage methodology approach was taken. The beginning point involved data collection from two major sources: U.S. Treasury yields (the 10-year yield and the 2-year yield, denoted as DGS10 and DGS2) and the recession indicator (USREC). These were downloaded from

FRED. S&P 500 index daily closing prices were downloaded simultaneously from Yahoo Finance. These data were merged on a common date index to be temporally aligned.

The second half included data preprocessing, wherein the yield spread was obtained as DGS10 minus DGS2. The dynamics of S&P 500 performance were represented through daily log returns. Since the data were time-series, observations with missing values were removed to maintain analytical precision.

Two primary statistical models were used. The initial model is a logistic regression, in which the binary recession indicator (USREC) is regressed upon the yield spread.

This was selected due to its appropriateness for binary outcomes, which allows us to measure the likelihood of a recession occurrence as a function of the behaviour of the yield.

The second model is a linear regression analysis, with S&P 500 returns modelled as a function of the lagged yield spread. This approach enables the testing of whether recent changes in the yield curve are likely to forecast short-horizon market performance. The use of lagged predictors ensures the absence of temporal causality and avoids look-ahead bias.

These methods were selected because they provide explanatory power and interpretability. Logistic regression provides direct information about recession probability, while linear regression on lagged variables provides actionable information for market timing strategies.

The overall data integration and clear model structure make the method robust and comprehensible to economists and finance practitioners.

Simulation

As a proof-of-concept for our analytical procedure, a simulation experiment was conducted.

In the simulation experiment, simulated artificial yield spread data were generated from a normal distribution. With a fixed logistic model having parameters $\beta_0 = -1$ and $\beta_1 = 4$, a binary recession indicator was constructed. Particularly, recession probability was calculated through the logistic transformation, and binary results were constructed by conducting a binomial draw.

This simulation serves two main purposes. First, it demonstrates the capability of the logistic regression model to recover the known parameters correctly when applied to data with well-

controlled statistical properties. Second, it provides a reference point against which the performance of the analysis on real data can be compared. The fitted logistic model on simulation data yielded parameter estimates that were near the actual values, thereby confirming that our methodology can reproduce the true relationship between the yield spread and recession probabilities.

By applying our methodology to simulated data, we are assured of the validity of our modeling method and its subsequent interpretation when applied to actual economic data. The simulation confirms the robustness of the logistic regression technique and its applicability in predicting binary economic responses from continuous predictors (Figure 1).

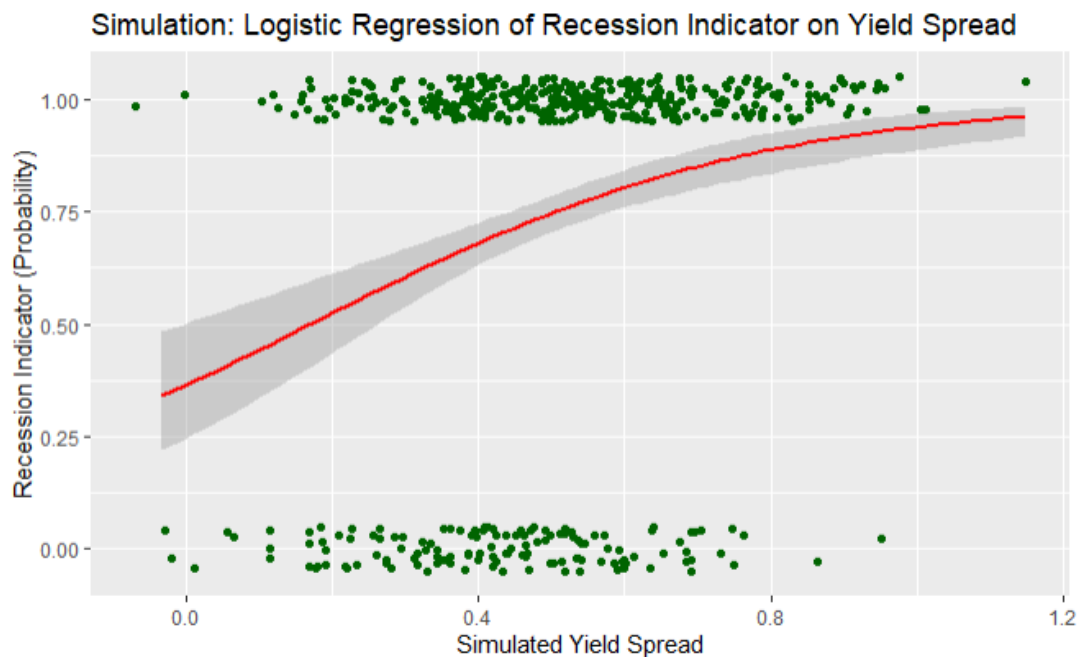


Figure 1

Real Data Analysis

The real analysis of the data was conducted using a combination of U.S. Treasury yield (DGS10 and DGS2), recession signals (USREC), and closing prices for the S&P 500. The datasets were synchronized and cleaned before the calculation of the yield spread, and this was mixed with daily S&P 500 log returns.

Two model specifications were calibrated on the cleaned data. In the logistic regression model, the recession indicator variable (USREC) was regressed on the yield spread. The model's output indicated a statistically significant relationship and supported the model's predictive ability in identifying recessions based on the yield curve (Figure 2). Graphical plots also amplified the time series pattern of the yield spread with a particular emphasis being on the overlapping episodes of inversions that fell within previous recessions.

Meanwhile, a linear regression model was run with the lagged yield spread as the independent variable for returns on the S&P 500. The findings provided a moderate yet statistically significant relationship, which demonstrated that short-run market performance can be explained by changes in the yield spread. A scatter plot of a fitted regression line provided a visual confirmation of this relationship, displaying the trend and corroborating the statistical findings (Figure 3).

The analysis steps, i.e., data acquisition, transformation, and model fitting, were carried out using R. Extended R code was executed to allow reproducibility and further plots were generated to present the results. This actual data analysis not only lends support to the theoretical justification provided by the simulation study but also provides valuable insight into how variations in the yield curve are expressed in market returns.

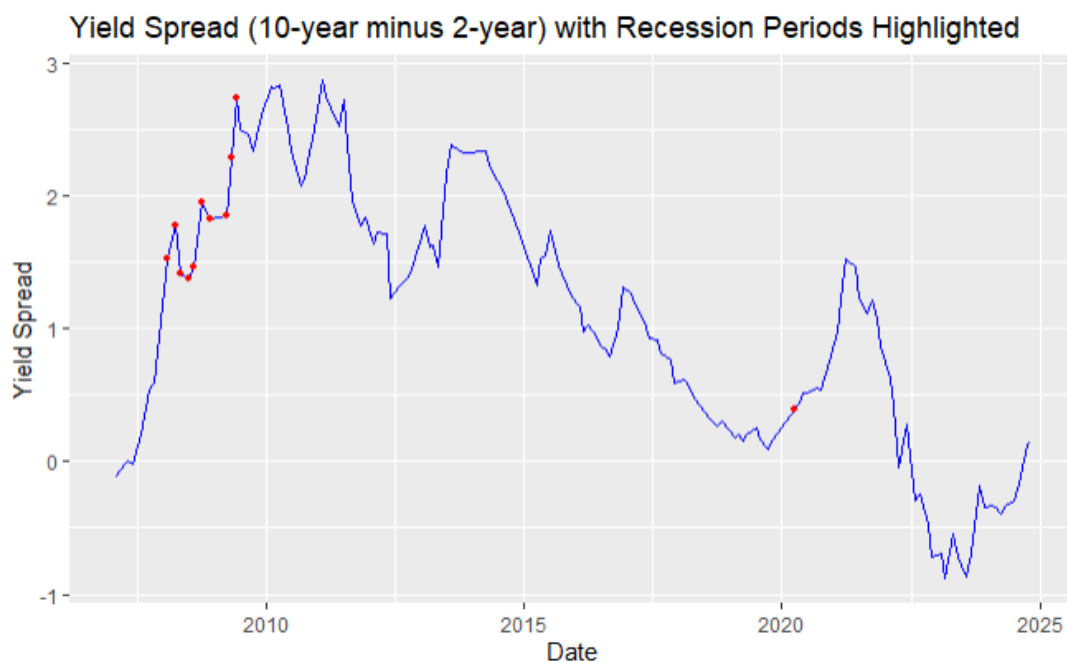


Figure 2

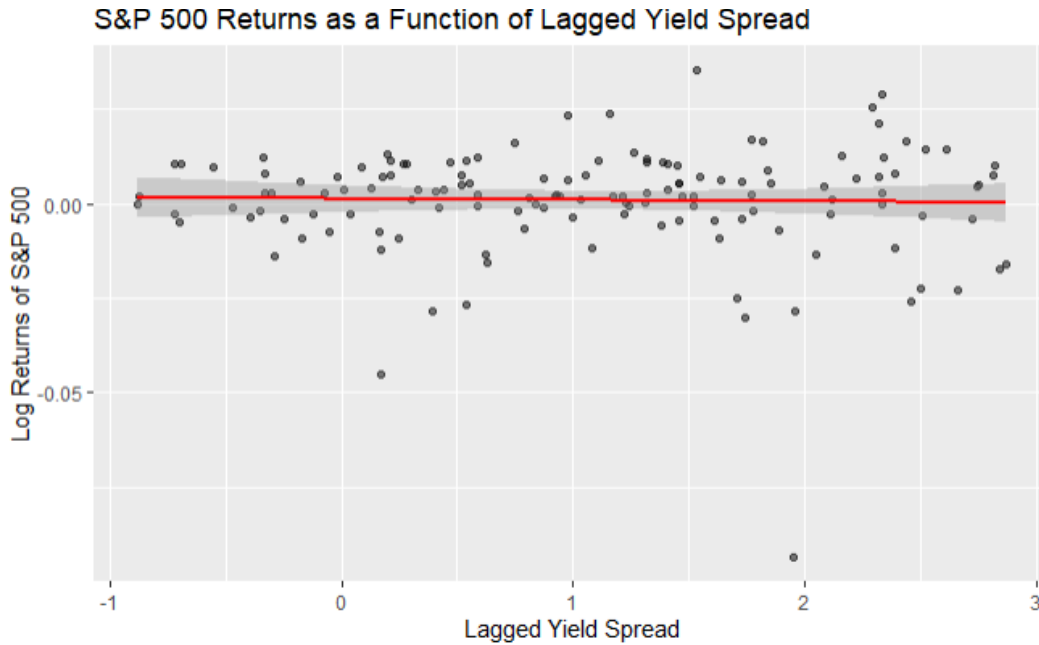


Figure 3

Conclusions and Discussion

This research demonstrates that the U.S. Treasury yield spread is a reliable indicator of economic recession and holds value in predicting S&P 500 return fluctuations. The logistic regression test confirms that yield curve inversions bear a strong correlation with subsequent recessionary periods. Further, the linear regression model indicates that yield spread fluctuations can be used to predict short-term market performance.

The findings have important practical implications. To policymakers, the yield curve provides an open early warning, which can allow for more forward-looking macroeconomic management. To financial analysts and investors, the results translate into a realistic tool to guide investment policy and risk management strategies. Finally, the successful simulation test enhances the robustness of the methods selected and justifies its use in subsequent studies. Subsequent studies can attempt to utilize additional macroeconomic variables to streamline and extend such findings.

References

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